

HIKMAON

Blockchain-Anchored AI Infrastructure
for Digital Authenticity & Deepfake Misuse Detection

Whitepaper

Muhammad Ayan Rao

Bestower Labs Limited

February 2026

1. Executive Technical Overview

Hikmaon is a real-time digital authenticity infrastructure system designed to:

- Automatically register original media at the moment of upload
- Anchor cryptographic proof of ownership on a proprietary hybrid blockchain (Hikmalayer)
- Continuously monitor publicly available internet content
- Detect similarity and AI-generated misuse using advanced artificial intelligence models
- Verify ownership through blockchain records
- Generate structured evidence with automated notification

Hikmaon is not merely a deepfake detection tool.

It is a full-stack authenticity infrastructure combining:

- Real-time media ingestion
- Cryptographic hashing
- Perceptual fingerprinting
- AI similarity embeddings
- Deepfake manipulation detection
- Public internet indexing
- Blockchain ownership anchoring
- Automated evidence generation
- Notification and reporting system

Development Status:

- Hikmalayer: Near completion (hybrid PoW + PoS operational)
- Hikmaon: Development started - advanced functional prototype / pre-production validation stage
- Core pipeline: Backend APIs for registration, connector onboarding, internet indexing, AI analysis, and incident management operational
- Patent: UK patent application filed covering core technology

2. Problem Statement

Artificial intelligence tools are now capable of:

- Generating highly realistic fake videos
- Cloning human voices
- Reconstructing faces
- Manipulating images convincingly
- Producing synthetic media at scale

When individuals upload media to social platforms or cloud storage:

- Ownership is not automatically registered
- There is no immutable proof of origin
- There is no automated internet-wide monitoring
- There is no structured detection of AI-generated misuse
- There is no integrated system that links AI detection with blockchain verification

Once media is online, it can be:

- Downloaded
- Modified
- Re-uploaded
- Used in AI models
- Misrepresented

The internet lacks built-in authenticity infrastructure.

Hikmaon is designed to solve this structural gap.

3. System Architecture Overview

Hikmaon operates through six primary system layers:

1. Account Integration Layer
2. Media Registration Engine
3. Blockchain Anchoring Layer (Hikmalayer)
4. AI Processing Layer (Three AI Systems)
5. Internet Discovery & Indexing Layer
6. Verification, Evidence & Notification Layer

3.1 Real-Time Social Media & Cloud Integration

Hikmaon connects to social media accounts and cloud storage services via secure authorised login (OAuth-based permission model).

When a user uploads new media:

- Platform webhook/API detects upload
- Hikmaon ingestion service receives metadata
- Media file (where permitted) is processed immediately
- Hash + fingerprint + embedding are generated
- Blockchain anchoring transaction is created
- Proof is written to Hikmalayer

This creates real-time ownership registration.

Important: The raw media is NOT stored on blockchain. Only digital representations are stored.

4. Technical Infrastructure & Patent-Aligned Diagrams

This section presents the detailed technical infrastructure of the Hikmaon system, illustrating the patent-pending workflows that establish digital media authenticity, detect misuse, and enforce rights through blockchain-anchored verification.

4.1 Overall System Architecture

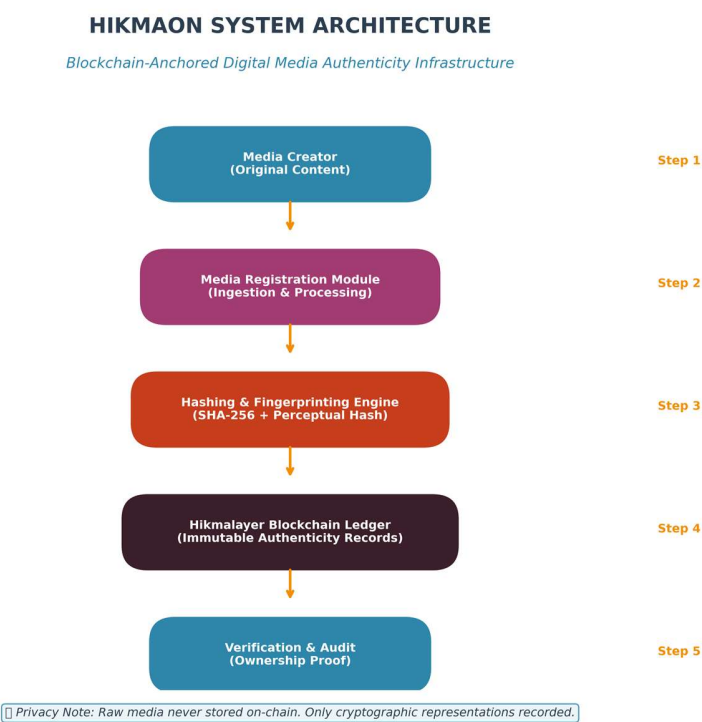


Figure 1: Hikmaon System Architecture - Blockchain-Anchored Digital Media Authenticity Infrastructure

The Hikmaon system operates through a five-layer vertical architecture:

1. **Media Creator Layer:** Original content generation by rights holders
2. **Media Registration Module:** Secure ingestion and processing gateway
3. **Hashing & Fingerprinting Engine:** Cryptographic representation generation (SHA-256 + perceptual hashing)
4. **Hikmalayer Blockchain Ledger:** Immutable authenticity record storage
5. **Verification & Audit Layer:** Ownership proof and forensic validation

Critical Privacy Feature: Raw media is never stored on the blockchain. Only derived cryptographic representations (hashes, fingerprints, and metadata) are recorded on-chain, ensuring complete privacy while maintaining verifiable authenticity.

4.2 Media Registration and Blockchain Anchoring Process

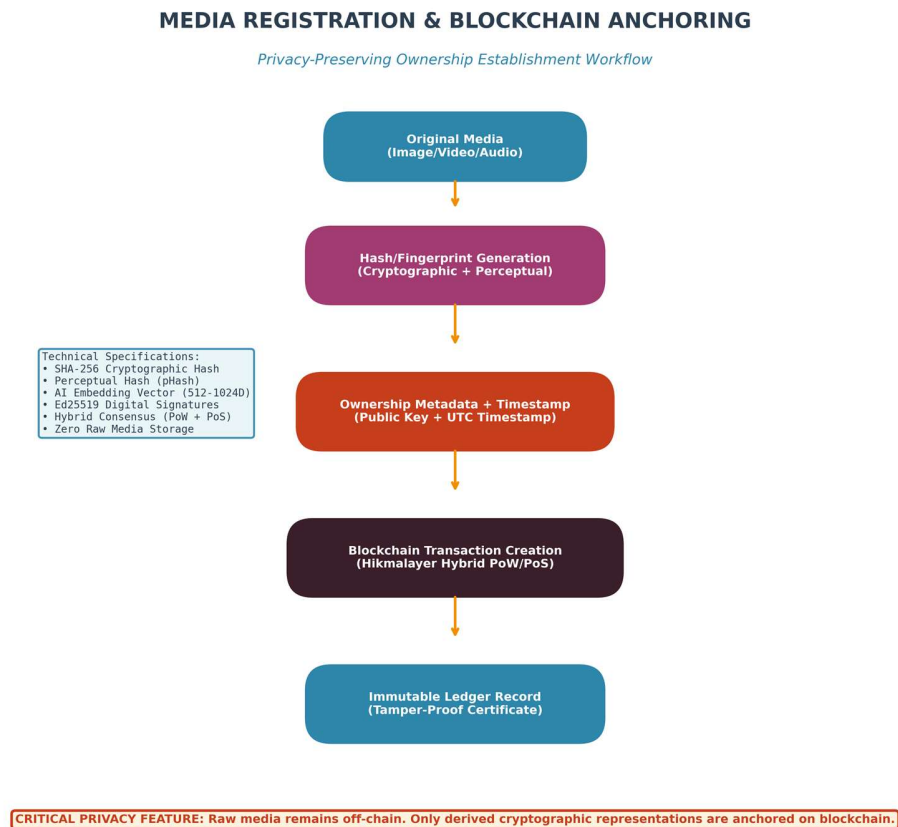


Figure 2: Privacy-Preserving Ownership Establishment Workflow

The registration workflow transforms original media into blockchain-anchored proof through five sequential stages:

Stage 1: Original Media Ingestion

- Supported formats: Image (JPEG, PNG, TIFF), Video (MP4, MOV, AVI), Audio (MP3, WAV, AAC)
- Secure upload via OAuth-authorized connectors or direct API submission

Stage 2: Hash/Fingerprint Generation

- **SHA-256 Cryptographic Hash:** Deterministic content identifier

- **Perceptual Hash (pHash):** Edit-resistant signature detecting visual similarity despite compression, resizing, or minor editing
- **AI Embedding Vector:** 512-1024 dimensional representation for semantic similarity search

Stage 3: Ownership Metadata Binding

- Owner public key (Ed25519 digital signature)
- UTC timestamp (nanosecond precision)
- Content type classification
- Optional licensing terms reference

Stage 4: Blockchain Transaction Creation

- Hikmalayer hybrid consensus (PoW + PoS)
- Structured transaction format: `MEDIA_REGISTRATION` type
- Merkle tree inclusion proof

Stage 5: Immutable Ledger Record

- Tamper-proof certificate generation
- Transaction ID assignment
- Global consensus confirmation

4.3 Deepfake and Media Misuse Detection Workflow

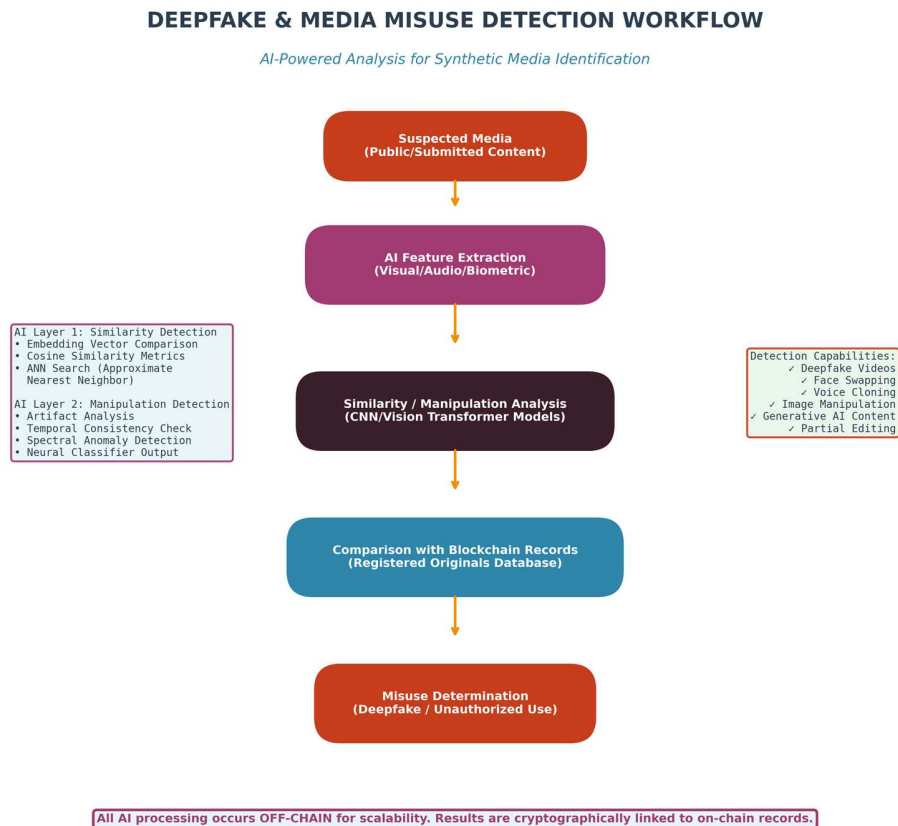


Figure 3: AI-Powered Analysis for Synthetic Media Identification

The detection engine operates through a three-layer AI architecture:

AI Layer 1: Similarity Detection

- Embedding vector comparison using cosine similarity metrics
- Approximate Nearest Neighbour (ANN) search for scalable matching
- Detection of cropped, resized, recompressed, or lightly edited derivatives

AI Layer 2: Manipulation Detection

- Visual artifact analysis (blending inconsistencies, lighting anomalies)
- Video temporal consistency checking
- Audio spectral anomaly detection
- Neural classifier outputting deepfake-probability score

AI Layer 3: Blockchain-Backed Verification

- Cross-reference against registered originals
- Cryptographic proof linking
- Confidence score computation

Detection Capabilities: Deepfake videos, face swapping, voice cloning, image manipulation, generative AI content, and partial editing.

4.4 Verification and Ownership Confirmation

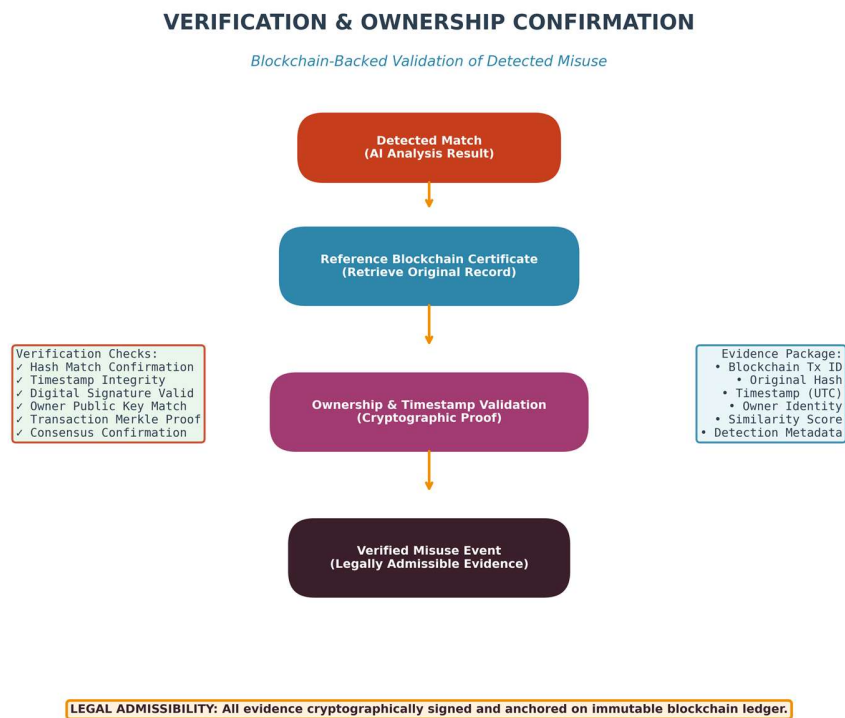


Figure 4: Blockchain-Backed Validation of Detected Misuse

Upon AI detection of potential misuse, the verification workflow ensures legal admissibility:

Verification Checks:

- Hash match confirmation between suspected and original media
- Timestamp integrity validation
- Digital signature verification
- Owner public key matching

- Transaction Merkle proof validation
- Consensus confirmation count

Evidence Package Generation:

- Blockchain transaction ID
- Original content hash
- UTC timestamp of original registration
- Owner identity (public key)
- Similarity score (percentage)
- Detection metadata (model version, analysis timestamp)

Legal Admissibility: All evidence is cryptographically signed and anchored on the immutable Hikmalayer blockchain ledger, providing court-admissible proof of ownership and timestamp.

4.5 Automated Notification and Rights Enforcement

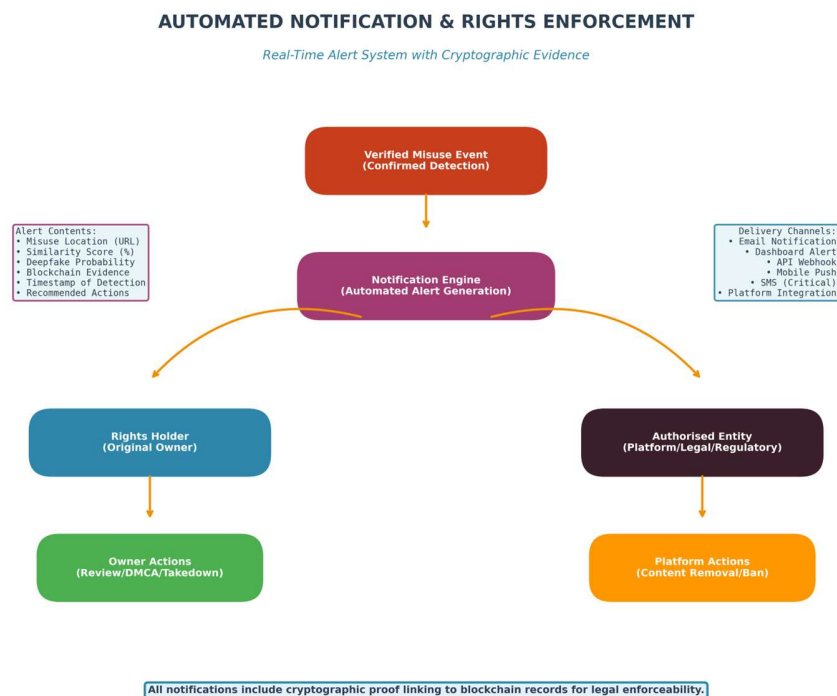


Figure 5: Real-Time Alert System with Cryptographic Evidence

The automated notification system triggers upon verified misuse detection:

Alert Contents:

- Misuse location (URL)
- Similarity score percentage
- Deepfake probability assessment
- Blockchain evidence reference
- Timestamp of detection
- Recommended enforcement actions

Delivery Channels:

- Email notification
- Dashboard alert
- API webhook for platform integration
- Mobile push notification
- SMS for critical alerts
- Direct platform integration

Recipient Actions:

- **Rights Holder:** Review evidence, file DMCA takedown, initiate legal proceedings
- **Authorised Entities:** Content removal, account suspension, regulatory reporting

All notifications include cryptographic proof linking to blockchain records for legal enforceability.

4.6 On-Chain and Off-Chain Architecture Separation

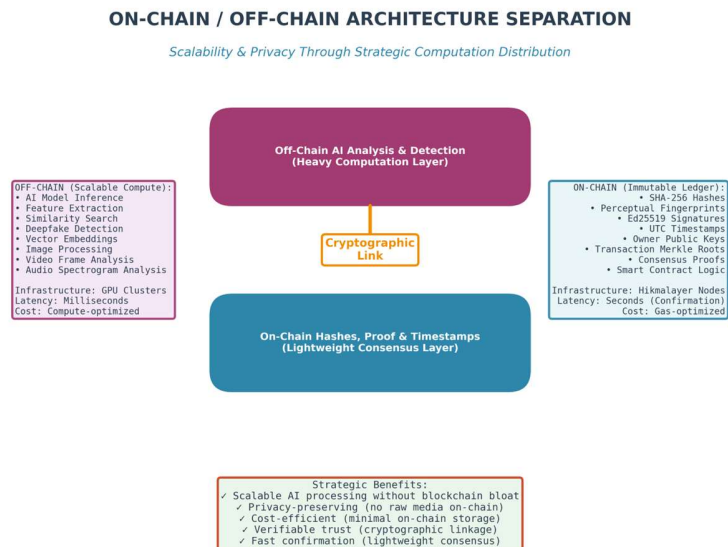


Figure 6: Scalability & Privacy Through Strategic Computation Distribution

Hikmaon employs a strategic separation between heavy computation and lightweight consensus:

Off-Chain Layer (Scalable Compute):

- AI model inference and training
- Feature extraction pipelines
- Similarity search operations
- Deepfake detection algorithms
- Vector embedding generation
- Image/video/audio processing

Infrastructure: GPU clusters | Latency: Milliseconds | Cost: Compute-optimized

On-Chain Layer (Immutable Ledger):

- SHA-256 hashes
- Perceptual fingerprints
- Ed25519 digital signatures
- UTC timestamps

- Owner public keys
- Transaction Merkle roots
- Consensus proofs

Infrastructure: Hikmalayer validator nodes | *Latency:* Seconds (confirmation time) |

Cost: Gas-optimized

Strategic Benefits:

- Scalable AI processing without blockchain bloat
- Privacy-preserving architecture (zero raw media on-chain)
- Cost-efficient operations (minimal on-chain storage)
- Verifiable trust through cryptographic linkage
- Fast confirmation via lightweight consensus

4.7 Complete End-to-End Workflow

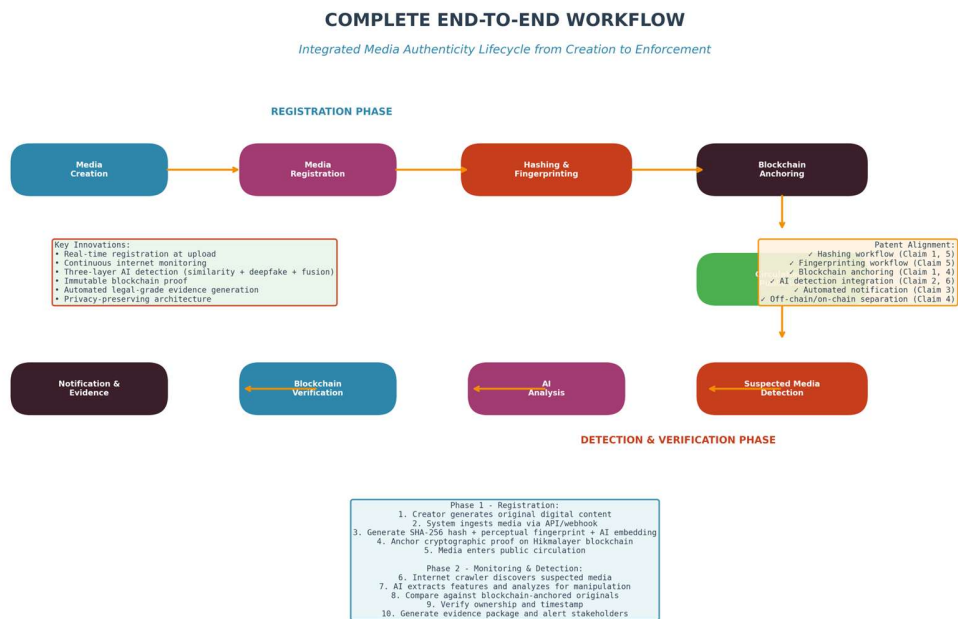


Figure 7: Integrated Media Authenticity Lifecycle from Creation to Enforcement

The complete Hikmaon workflow integrates two major phases:

Phase 1: Registration (Proactive Protection)

1. Creator generates original digital content

2. System ingests media via secure API/webhook
3. Generate SHA-256 hash + perceptual fingerprint + AI embedding
4. Anchor cryptographic proof on Hikmalayer blockchain
5. Media enters public circulation with authenticity certificate

Phase 2: Monitoring & Detection (Reactive Enforcement)

1. Internet crawler continuously discovers suspected media
2. AI extracts features and analyzes for manipulation/synthesis
3. Compare against blockchain-anchored original records
4. Verify ownership and timestamp through cryptographic proof
5. Generate evidence package and alert stakeholders

Key Innovations:

- Real-time registration at moment of upload
- Continuous internet-wide monitoring
- Three-layer AI detection (similarity + deepfake + fusion)
- Immutable blockchain proof of authenticity
- Automated legal-grade evidence generation
- Privacy-preserving architecture

Patent Alignment: This workflow directly implements the patent claims filed:

- Claim 1: Hashing and blockchain anchoring method
- Claim 2: AI detection system with blockchain records
- Claim 3: Automated notification method
- Claim 4: On-chain/off-chain separation framework
- Claim 5: Cryptographic hashes and perceptual fingerprints
- Claim 6: AI analysis of visual/audio/biometric features

5. Media Registration Workflow (Patent-Aligned)

Upon upload:

Step 1 – Cryptographic Hash

Generate SHA-256 hash of media file.

Step 2 – Perceptual Fingerprint

Generate edit-resistant signature:

- Image: perceptual hash
- Video: frame sampling fingerprint
- Audio: acoustic feature fingerprint

Step 3 – AI Embedding Vector

Generate high-dimensional embedding representation for similarity search.

Step 4 – Blockchain Transaction

Create structured transaction including:

- content_hash
- fingerprint_commitment
- owner_public_key
- timestamp
- metadata_reference

Submit to Hikmlayer.

Step 5 – Store Off-Chain Records

Embeddings and fingerprints stored in secure database.

This structure matches patent filing elements: hashing → fingerprinting → blockchain anchoring → AI verification → notification.

6. Hikmalayer Hybrid Blockchain

Hikmalayer is a hybrid consensus blockchain combining:

- Proof of Work (security)
- Proof of Stake (efficiency and governance)

Reasons for building proprietary hybrid chain instead of using Bitcoin/Ethereum:

- Purpose-built transaction structure for media authenticity
- Controlled transaction cost and scalability
- Faster confirmation for high-volume media registration
- Governance control aligned with authenticity roadmap
- Patent-aligned blockchain architecture
- Elimination of dependency on external chain upgrades

Bitcoin is monetary-focused. Ethereum is smart-contract-focused. **Hikmalayer is authenticity-focused.**

7. Internet Monitoring & Discovery Layer

Hikmaon operates a public media discovery engine.

It functions similarly to a search engine but indexes media fingerprints rather than text.

Components:

- Public Web Crawler (lawful, respecting robots.txt)
- Media Extractor
- Lightweight Fingerprint Generator
- Fingerprint Index Database
- Approximate Nearest Neighbor (ANN) Search Engine

Workflow:

1. Crawl public pages
2. Extract media
3. Generate lightweight fingerprint
4. Store URL + fingerprint in index
5. Compare against registered originals

When similarity threshold is exceeded: Deep AI analysis is triggered.

This allows detection of:

- Past misuse
- Present misuse
- Future uploads

Note: Monitoring is public-content based. Private content is accessed only via authorised integrations.

8. Three-Layer AI Architecture

Hikmaon uses three AI layers.

8.1 AI Layer 1 – Similarity Embedding System

Purpose: Detect derived or similar content across internet.

Method:

Media → AI encoder → embedding vector (512–1024 dimensions). Similarity measured using cosine similarity.

Technologies:

- CNN or Vision Transformer for images
- Frame-based encoding for video
- Spectrogram-based encoder for audio
- ANN search (vector index)

This allows detection even if media is:

- Cropped
- Resized
- Recompressed
- Slightly edited

8.2 AI Layer 2 – Deepfake & Manipulation Detection

Purpose: Detect synthetic or manipulated content.

Methods include:

- Image artifact detection (blending inconsistencies, lighting anomalies)
- Video temporal inconsistency analysis
- Audio spectral anomaly detection
- Neural classifier outputs probability of manipulation

Outputs: deepfake_probability score.

8.3 AI Layer 3 – Decision Fusion Engine

Purpose: Combine detection signals with blockchain verification.

Inputs:

- similarity_score
- deepfake_probability
- blockchain_verified_flag

Decision logic: Compute final confidence score.

If threshold exceeded: Trigger evidence generation + alert.

This ensures AI result is backed by blockchain proof.

9. How AI Is Achieved (Practical Implementation)

AI is implemented using:

- Pretrained encoder models for embeddings
- Fine-tuned classifiers for manipulation detection
- Vector search engine for scalable matching
- Structured inference pipeline for performance

Heavy AI runs off-chain. Blockchain remains lightweight and proof-focused.

Training Strategy:

1. Start with pretrained models
2. Fine-tune on curated deepfake datasets
3. Gradually improve using collected misuse samples

10. Verification & Evidence System

When potential misuse detected:

1. Retrieve original registration
2. Query Hikmalayer for transaction
3. Confirm hash + owner + timestamp
4. Generate evidence pack including:
 - blockchain transaction ID
 - similarity score
 - deepfake probability
 - URL of misuse
 - timestamp of detection
 - model version metadata
5. Notify user

Optional reporting tools assist in complaint submission.

11. Patent Alignment Summary

Filed patent covers:

- Hashing workflow
- Fingerprinting workflow
- Blockchain anchoring structure
- AI similarity & manipulation detection integration
- Automated notification system
- Evidence generation architecture

Hikmaon implementation strictly follows this structure.

12. Strategic Positioning

Hikmaon is not just detection software.

It is:

AI-powered authenticity infrastructure anchored on proprietary hybrid blockchain.

It connects:

Real-time media upload → Blockchain proof → Internet-wide monitoring → AI detection → Verified ownership → Evidence → Alert

System Flow Diagram:

1. USER UPLOADS MEDIA
2. → HASH + FINGERPRINT + EMBEDDING GENERATED
3. → BLOCKCHAIN TRANSACTION CREATED
4. → PROOF ANCHORED ON HIKMALAYER
5. INTERNET MONITORING (Continuous)
6. → SIMILARITY DETECTION → AI ANALYSIS
7. → OWNERSHIP VERIFICATION
8. → EVIDENCE GENERATION → USER ALERT